

## TECUCI, Romania

WMO: 152650

Lat: 45.8417N

Lon: 27.4089E

Elev: 201

StdP: 14.59

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 9.1        | 13.3 | 4.0                           | 6.8 | 11.1 | 7.6 | 8.2 | 14.8 | 28.8                  | 23.2 | 25.4 | 22.3 | 5.4                   | 340  | 0.493 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 23.1                   | 91.6            | 69.9 | 89.0 | 69.3 | 86.6 | 68.7 | 73.2                | 86.1 | 71.5 | 84.4 | 70.2 | 82.8 | 6.0                  | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.0                            | 107.4 | 79.2 | 67.4 | 101.3 | 76.3 | 66.0 | 96.7 | 74.8 | 36.8          | 86.3 | 35.3 | 84.9 | 34.2 | 83.5 | 79.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 21.4              | 18.5 | 16.2 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 52.3   | 27.0  | 35.2  | 42.2  | 51.7  | 63.1  | 70.6  | 72.2  | 72.9  | 64.2  | 51.8  | 42.9  | 32.3 |
|                                                                               | DBStd       | 17.20  | 7.92  | 8.71  | 8.28  | 7.85  | 5.79  | 5.15  | 4.51  | 5.43  | 7.38  | 7.31  | 7.66  | 8.02 |
|                                                                               | HDD50       | 2321   | 713   | 415   | 260   | 76    | 3     | 0     | 0     | 0     | 1     | 67    | 238   | 548  |
|                                                                               | HDD65       | 5435   | 1178  | 833   | 707   | 400   | 102   | 12    | 4     | 7     | 105   | 410   | 664   | 1013 |
|                                                                               | CDD50       | 3148   | 0     | 2     | 18    | 129   | 410   | 618   | 689   | 709   | 426   | 123   | 24    | 0    |
|                                                                               | CDD65       | 787    | 0     | 0     | 0     | 2     | 44    | 180   | 228   | 250   | 81    | 2     | 0     | 0    |
|                                                                               | CDH74       | 7748   | 0     | 0     | 6     | 52    | 534   | 1642  | 2123  | 2558  | 812   | 21    | 0     | 0    |
|                                                                               | CDH80       | 2832   | 0     | 0     | 0     | 3     | 111   | 560   | 788   | 1119  | 251   | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 6.7    | 7.0   | 7.0   | 8.1   | 7.8   | 6.8   | 6.8   | 6.1   | 6.4   | 5.8   | 5.7   | 6.0   | 6.6  |
| Precipitation                                                                 | PrecAvg     | 21.3   | 1.0   | 0.9   | 1.1   | 1.8   | 2.4   | 2.9   | 2.4   | 2.0   | 1.9   | 2.0   | 1.3   | 1.4  |
|                                                                               | PrecMax     | 31.0   | 2.1   | 3.1   | 2.7   | 3.6   | 5.4   | 6.9   | 6.7   | 6.2   | 5.6   | 8.1   | 4.3   | 4.6  |
|                                                                               | PrecMin     | 12.3   | 0.1   | 0.1   | 0.1   | 0.6   | 0.3   | 0.4   | 0.4   | 0.5   | 0.2   | 0.1   | 0.0   | 0.0  |
|                                                                               | PrecStd     | 5.1    | 0.7   | 0.7   | 0.7   | 0.9   | 1.4   | 1.7   | 1.5   | 1.4   | 1.3   | 1.7   | 1.0   | 1.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 48.4  | 62.0  | 72.2  | 79.1  | 88.0  | 92.7  | 93.5  | 94.7  | 89.6  | 77.6  | 68.0  | 55.0 |
|                                                                               |             | MCWB   | 41.6  | 50.7  | 54.6  | 62.0  | 66.7  | 72.4  | 71.4  | 71.1  | 66.6  | 63.4  | 57.9  | 49.0 |
|                                                                               | 2%          | DB     | 43.0  | 55.6  | 64.6  | 75.0  | 83.2  | 89.0  | 90.4  | 92.2  | 86.0  | 72.7  | 62.8  | 50.4 |
|                                                                               |             | MCWB   | 39.9  | 47.7  | 51.0  | 59.2  | 63.9  | 70.7  | 70.5  | 69.4  | 66.3  | 60.4  | 54.9  | 45.9 |
|                                                                               | 5%          | DB     | 40.0  | 51.5  | 60.2  | 70.6  | 79.9  | 86.2  | 87.7  | 89.5  | 82.7  | 68.4  | 57.4  | 46.0 |
|                                                                               |             | MCWB   | 37.7  | 44.8  | 48.9  | 56.9  | 63.4  | 69.9  | 70.0  | 68.9  | 64.3  | 58.2  | 51.5  | 42.8 |
|                                                                               | 10%         | DB     | 36.9  | 47.1  | 55.6  | 66.8  | 76.8  | 83.1  | 84.6  | 86.7  | 78.6  | 64.2  | 53.6  | 41.9 |
|                                                                               |             | MCWB   | 34.9  | 42.5  | 46.2  | 54.7  | 61.7  | 68.1  | 68.9  | 68.2  | 63.5  | 55.8  | 50.0  | 39.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 43.4  | 53.6  | 56.1  | 63.0  | 68.6  | 75.7  | 74.6  | 76.1  | 70.4  | 65.1  | 58.4  | 51.0 |
|                                                                               |             | MCDB   | 46.3  | 60.2  | 69.9  | 76.9  | 83.9  | 87.8  | 88.9  | 91.4  | 83.3  | 74.8  | 66.1  | 52.8 |
|                                                                               | 2%          | WB     | 40.5  | 48.0  | 52.2  | 60.3  | 66.6  | 73.4  | 72.5  | 72.5  | 67.8  | 61.7  | 55.6  | 46.8 |
|                                                                               |             | MCDB   | 42.6  | 53.4  | 63.2  | 72.7  | 79.0  | 85.0  | 85.7  | 86.5  | 80.0  | 69.7  | 61.7  | 50.1 |
|                                                                               | 5%          | WB     | 37.8  | 45.1  | 49.5  | 58.1  | 64.6  | 71.2  | 70.9  | 71.0  | 66.4  | 59.3  | 53.0  | 43.1 |
|                                                                               |             | MCDB   | 39.8  | 49.9  | 57.9  | 68.6  | 76.1  | 82.6  | 83.8  | 84.8  | 78.3  | 66.2  | 56.8  | 46.2 |
|                                                                               | 10%         | WB     | 35.2  | 43.0  | 47.4  | 55.8  | 62.6  | 69.5  | 69.7  | 69.5  | 64.6  | 57.2  | 50.2  | 39.4 |
|                                                                               |             | MCDB   | 36.9  | 46.7  | 54.2  | 64.9  | 73.2  | 80.6  | 82.5  | 82.3  | 76.5  | 62.6  | 53.2  | 41.6 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 11.0  | 13.6  | 17.9  | 19.8  | 21.6  | 20.4  | 21.1  | 23.1  | 21.5  | 18.0  | 12.7  | 10.3 |
|                                                                               |             | MCDBR  | 13.9  | 22.2  | 27.9  | 27.9  | 27.5  | 24.9  | 26.4  | 27.6  | 28.3  | 25.3  | 20.5  | 15.8 |
|                                                                               |             | MCWBR  | 10.4  | 14.6  | 15.5  | 14.7  | 11.4  | 10.0  | 9.5   | 9.5   | 11.3  | 13.6  | 12.4  | 11.5 |
|                                                                               | 5% WB       | MCDBR  | 12.4  | 19.0  | 24.6  | 26.4  | 23.7  | 22.6  | 23.8  | 24.8  | 25.1  | 22.9  | 18.3  | 15.5 |
|                                                                               | MCWBR       | 9.7    | 13.4  | 14.7  | 14.1  | 10.8  | 9.8   | 9.3   | 9.2   | 10.9  | 13.0  | 11.7  | 11.6  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.331  | 0.362 | 0.397 | 0.457 | 0.429 | 0.428 | 0.447 | 0.460 | 0.418 | 0.383 | 0.357 | 0.331 |      |
|                                                                               | taud        | 2.312  | 2.245 | 2.198 | 2.055 | 2.217 | 2.256 | 2.211 | 2.162 | 2.243 | 2.321 | 2.344 | 2.326 |      |
|                                                                               | Ebn at Noon | 239    | 253   | 260   | 254   | 267   | 267   | 260   | 251   | 251   | 243   | 227   | 225   |      |
|                                                                               | Edh at Noon | 25     | 33    | 39    | 49    | 44    | 42    | 44    | 44    | 37    | 30    | 24    | 23    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 410    | 664   | 1032  | 1405  | 1780  | 1880  | 1912  | 1732  | 1250  | 810   | 441   | 338   |      |
|                                                                               | RadStd      | 50     | 79    | 106   | 139   | 149   | 151   | 148   | 123   | 117   | 92    | 41    | 40    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (58 neighbors) | +1.03 | N/A     | N/A    | N/A     | N/A   | N/A   | -207        | -279  | +131  | +52   |

Nomenclature: See separate page